

CyberOps Associate v1.0

MALWARE INVESTIGATION

SECURITY ONION V16.04

Hands-On Skills Assessment: Pushdo Trojan Forensic Analysis

Date: October 2026 | Score: 100%

Scenario Overview & Assessment Scope

You are acting as a junior security analyst in a Security Operations Center (SOC). As part of incident response training, you were tasked with determining malicious activity associated with the **Pushdo Trojan** on the company network using a **Security Onion** monitoring environment (Sguil, Kibana, NetworkMiner, and Wireshark) alongside external threat intelligence (VirusTotal).

Target Subject: Pushdo (Downloader/Cutwall botnet) **Incident Timeline:** 2017-06-27 **Primary Tools:** Sguil, NetworkMiner, tshark, VirusTotal

PART 1: GATHER THE BASIC INFORMATION

Step 1: Verify the status of services

- a. Log into Security Onion VM with username **analyst** and password **cyberops**.
- b. Open a terminal window. Enter the **sudo so-status** command to verify that all services are ready.

```
analyst@SecOnion:~$ sudo so-status
Status: securityonion
  * sguil server [ OK ]
Status: seconion-import
  * pcap_agent (sguil) [ OK ]
  * snort_agent-1 (sguil) [ OK ]
  * barnyard2-1 (spooler, unified2 format) [ OK ]
Status: Elastic stack
  * so-elasticsearch [ OK ]
  * so-logstash [ OK ]
  * so-kibana [ OK ]
  * so-freqserver [ OK ]
```

- c. Launch Sguil or Kibana using credentials **analyst / cyberops**. Click **Select All** to select sensor interfaces and click **Start SGUIL**.

Step 2: Gather basic information

- a. Identify the time frame of the Pushdo trojan attack, including the date and approximate time.

2017-06-27 from 13:38:34 to 13:44:32 UTC

The initial HTTP GET payload request occurs at **13:38:32 UTC**. The first NIDS alert triggers at **13:38:34 UTC**, and the final Tor SSL alert concludes at **13:44:32 UTC**.

- b. List the alerts noted during this time frame associated with the trojan.

- ET CURRENT_EVENTS WinHttpRequest Downloading EXE
- ET POLICY PE EXE or DLL Windows file download HTTP
- ET CURRENT_EVENTS Terse alphanumeric executable downloader high likelihood of being hostile
- ET POLICY External IP Lookup Domain (myip.opendns .com in DNS lookup)
- ET TROJAN Backdoor.Win32.Pushdo.s Checkin
- ET TROJAN Pushdo.S CnC response
- ET POLICY TLS possible TOR SSL traffic

c. List the internal IP addresses and external IP addresses involved.**Internal IP Address (Victim Host):**

- **192.168.1.96** (Infected Windows Workstation)

External IP Addresses (Infrastructure & C2):

- **119.28.70.207** — Serves **matied.com** (**/gerv.gun**) & **centler.at**
- **145.131.10.21** — Serves **lounge-haarstudio.nl** (**/oud/trow.exe**)
- **143.95.151.192** — Serves **vantagepointtechnologies.com** (**/wp.exe**)
- **208.67.222.222** — OpenDNS Resolver (DNS query for **myip.opendns.com**)
- **198.1.85.250** — Pushdo Trojan C2 Check-in Server (Port 80)
- **62.210.140.158** — Pushdo C2 Command & Control Response Server
- **208.83.223.34** — External Tor SSL / TLS endpoint

PART 2: LEARN ABOUT THE EXPLOIT

Step 1: Infected Host

a. Based on the alerts, what is the IP and MAC addresses of the infected computer? Based on the MAC address, what is the vendor of the NIC chipset?

IP Address: 192.168.1.96

MAC Address: 00:15:C5:DE:C7:3B

NIC Vendor: Dell Inc. (OUI 00:15:C5 belongs to Dell Inc.)

(Host Name identified via NetBIOS/DHCP packets: *FlashGordon-PC*)

Verification Method: In Sguil, right-click Alert ID 5410 → Select **NetworkMiner** → Open the **Hosts** tab and inspect the 192.168.1.96 entry.

b. Based on the alerts, when (date and time in UTC) and how was the PC infected? How did the malware infect the PC?

Infection Timestamp: 2017-06-27 13:38:32 UTC

Infection Vector & Mechanism:

The user on 192.168.1.96 initiated an HTTP connection via the Windows **WinHttpRequest** library to **matied.com/gerv.gun**. The downloaded payload **gerv.gun** was a compiled Windows 32-bit PE executable masquerading under a non-standard file extension.

Once executed, **Pushdo** functioned as a **downloader trojan**. Pushdo reports back to hardcoded command-and-control (C2) servers over TCP port 80, masquerading as an Apache web server. Before requesting additional malware, Pushdo harvests detailed host intelligence, including:

- Victim user administrative privilege status.
- Primary hard drive hardware serial number (queried via **SMART_RCV_DRIVE_DATA** IO control code).
- Filesystem type check (determining if the system is NTFS).
- Windows operating system version (retrieved via the **GetVersionEx** Win32 API call).
- Execution counter tracking how many times Pushdo has run on the endpoint.

Step 2: Examine the Exploit

a. Based on the alerts associated with HTTP GET requests, what files were downloaded? List the malicious domains observed, files downloaded, and their SHA256 hashes.

FILE NAME	DOMAIN & URI	SHA-256 HASH	FILE TYPE
gerv.gun	matied.com/gerv.gun	0931537889c35226d00ed26962ecacb140521394279eb2ade7e9d2afc1a7272	PE32 GUI Executable (Intel 386, 236 KB)
trow.exe	lounge-haarstudio.nl/oud/trow.exe	94a0a09ee6a21526ac34d41eabf4ba603e9a30c26e6a1dc072ff45749dfb1fe1	PE32 GUI Executable (Intel 386, 323 KB)
wp.exe	vantagepointtechnologies.com/wp.exe	79d503165d32176842fe386d96c04fb70f6ce1c8a485837957849297e625ea48	PE32 GUI Executable (Intel 386, 300.5 KB)

Method: In NetworkMiner, navigate to the **Files** tab, right-click the extracted file, and select **Calculate MD5 / SHA1 / SHA256 hash** (or run **sha256sum** on the extracted bro objects).

b. VirusTotal Threat Intelligence Verification**1. Payload: gerv.gun (SHA-256: 09315378...)****58 SECURITY ENGINES FLAGGED****Classification:** Win32.Trojan.Pushdo / Trojan:Win32/Wacatac | **Size:** 236.00 KB (241,664 bytes)**Known Aliases:** test , tmp523799.697 , vector.tui , extract-1498570714.111294-HTTP-FG0jno3bJLiIzR4hrh.exe**Target Architecture:** Intel 386 or later compatible 32-bit Windows subsystem.**2. Payload: throw.exe (SHA-256: 94a0a09e...)****63 SECURITY ENGINES FLAGGED****Classification:** Win32.Trojan.Cutwail / Trojan.GenericKD | **Size:** 323.00 KB (330,752 bytes)**Known Aliases:** Pedals.exe , throw.exe , test3 , bma2beo4.exe**Behavior:** Secondary botnet spam component associated with Cutwail payload delivery.**3. Payload: wp.exe (SHA-256: 79d50316...)****55 SECURITY ENGINES FLAGGED****Classification:** Win32.Malware-gen / Trojan.Downloader | **Size:** 300.50 KB (307,712 bytes)**Known Aliases:** wp.exe , test2 , test_3 , 4da48f6423d5f7d75de281a674c2e620.viobj**c. Examine other alerts associated with the infected host during this timeframe and record your findings.****1. External IP Lookup Alert:** ET POLICY External IP Lookup Domain (myip.opendns.com in DNS lookup) — Target: 208.67.222.222:53 .**Analysis:** myip.opendns.com is a legitimate OpenDNS IP reflection utility. Malware intentionally queries this domain to detect its public WAN address to report the host's geographical position to the C2 server.**2. C2 Obfuscation Flooding (Decoy Traffic):** Immediately following the Pushdo check-in (at 13:44:01), the host sends a barrage of HTTP GET requests to hundreds of benign, legitimate websites (e.g., vivastreet.com , hazmatt.com , kursavto.ru , themark.org). This is a known Pushdo evasion technique designed to create traffic noise and camouflage genuine C2 callbacks.**3. Encrypted Egress:** Alert ET POLICY TLS possible TOR SSL traffic directed to 208.83.223.34 indicates an attempt to establish an encrypted C2 or exfiltration tunnel.**Step 3: Report Your Findings (Executive Incident Report)****SOC Incident Response Executive Summary****Incident Narrative:** On June 27, 2017, between 13:38:34 and 13:44:32 UTC, internal workstation 192.168.1.96 (hostname FlashGordon-PC , MAC 00:15:C5:DE:C7:3B , Dell Inc.) was compromised by the Pushdo Trojan downloader.

The initial compromise occurred when the user navigated to matied.com/gerv.gun , downloading an obfuscated Win32 executable. Upon execution, Pushdo collected local hardware/OS telemetry, discovered its external WAN IP via OpenDNS (208.67.222.222), and established C2 communications with 198.1.85.250 and 62.210.140.158 over port 80.

Pushdo subsequently retrieved two secondary malicious binaries: throw.exe (from lounge-haarstudio.nl) and wp.exe (from vantagepointtechnologies.com). All three payloads were confirmed as critical threats on VirusTotal (55+ detections each). To hinder network detection, the malware flooded the network with rapid HTTP GET requests to random benign sites while initiating encrypted Tor SSL communication with 208.83.223.34 .

Remediation Actions Taken: (1) Isolate host 192.168.1.96 from the local subnet; (2) Block all discovered external malicious IPs and domains on the perimeter firewall and DNS proxy; (3) Capture volatile forensic memory and re-image the host endpoint.